DESCRIPTIVE ANALYTICS

US POLO SHOES

SALES • CUSTOMER PREFERENCES • PRODUCT PERFORMANCE

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Course	Bachelor of Computer Applications (BCA) — Data Science & AI
Subject	Descriptive Analytics / Data Visualization
Dataset	US Polo Shoes Sales Transaction Dataset
Analysis Period	1 January 2025 – 1 May 2025

DATA-DRIVEN FOOTWEAR RETAIL INSIGHTS

Academic disclaimer: All transaction-level findings are generated from an academic project dataset for educational purposes. They are not official U.S. Polo Assn. / USPA Global Licensing sales, customer, revenue, market-share, or product-performance statistics.

Table of Contents

Cover Page 1

Table of Contents 2

Keywords 3

Note on Data 4

Objectives 5

How This Would Work at Real Scale 6

Analytics Stack 7

1. Top-Selling Shoe Models 8

2. Monthly Sales Trend 10

3. Average Order Value & Distribution 11

4. Size Tier Allocation 12

5. Material Type Share 13

6. Regional Demand Distribution 14

7. Revenue Concentration 15

8. Customer Payment Preferences 16

9. Price & Basket Size Relationships 17

Summary of Key Metrics 18

Next Steps: Descriptive → Predictive Analytics 19

References 20

Appendix 21

Keywords

Keywords: US Polo Shoes, Descriptive Analytics, Data Visualization, Sales Analysis, Customer Preferences, Product Performance, Footwear Retail Analytics, Academic Project Dataset, Python, Pandas, Jupyter Notebook, Matplotlib, Data Cleaning, Sales Trend, Average Order Value, Size Tier Allocation, Material Type Share, Regional Demand, Payment Preferences, Revenue Concentration, Key Performance Indicators (KPIs), Business Insights, Predictive Analytics, Retail Analytics, Inventory Planning, Customer Segmentation.

Key Terms Used in the Report

Descriptive Analytics: The process of organizing, summarizing and interpreting historical or observed data to understand what happened.

Data Visualization: Representing numerical information through charts and graphs to make trends, comparisons and patterns easier to identify.

Product Performance: A measure of how individual shoe models perform using indicators such as units sold, revenue and average order value.

Customer Preference: The observed tendency of customers toward particular models, sizes, materials, regions or price levels.

Academic Project Dataset: Transaction data compiled for academic descriptive-analytics coursework, resembling realistic retail records.

Business Insight: A practical interpretation of an analytical result that can support inventory, merchandising or marketing decisions.

KPI: A key performance indicator that summarizes an important business measure.

Predictive Analytics: A future analytical stage in which historical data are used to estimate future demand or customer behaviour.

Analytical Scope

The report focuses on US Polo footwear sales and customer purchasing behaviour. The analytical approach follows the sample report format by presenting the figure context, observation, business insight and measurable KPIs for each major topic.

Note on Data

This report uses the US Polo Shoes Sales Transaction Dataset compiled for academic descriptive analytics. It is not a company-owned or officially published U.S. Polo Assn. / USPA Global Licensing transaction database.

The dataset contains 100 order records spanning a simulated period from 1 January 2025 to 1 May 2025, representing 355 total pairs sold and $25,879.10 USD in gross revenue. It includes product information, customer geography, size and material attributes, pricing, quantity, sale date and payment method.

Countries covered: Orders in the dataset ship to six destination markets — United Kingdom, USA, India, Mexico, Pakistan and Brazil — with each transaction recording a Country field (delivery destination market) and a Region field (state/regional detail within that country). The United Kingdom is the highest-revenue single market in the dataset, followed by the USA; all monetary figures throughout this report are expressed in USD regardless of destination market.

Because May 2025 contains only 5 orders logged on a single day (a collection-boundary truncation), monthly figures for that period should be read as partial rather than a genuine demand decline.

Dataset Variables

- Order_ID — Unique transaction identifier
- Customer_Name — Anonymized customer identifier
- Country — Delivery destination market
- Region — Delivery region/state within the destination country
- Shoe_Model — Footwear model (6 distinct models)
- Size_Tier — UK 6-8, UK 9-10, or UK 11-12
- Material — Upper material format (5 distinct types)
- Quantity — Pairs purchased per order
- Price — Unit selling price (USD)
- Sale_Date — Transaction date
- Payment_Method — Settlement channel
- Revenue — Quantity × Price

Objectives

The main purpose of this descriptive analytics project is to understand sales performance and customer preferences associated with US Polo footwear. The project converts raw transaction records into understandable charts, summary measures and business insights.

Q.1 Which shoe models sell the most by revenue and by physical volume?

Q.2 How does sales activity vary from month to month, and which periods show the highest and lowest revenue?

Q.3 What is the Average Order Value (AOV), and how are transaction sizes distributed?

Q.4 How does demand differ across size tiers, and what does this mean for inventory allocation?

Q.5 Which upper-material types contribute the most to revenue?

Q.6 How are sales distributed across geographic markets, and where is demand most concentrated?

Q.7 How concentrated is revenue among the leading shoe models?

Q.8 Which payment methods do customers prefer?

Q.9 How are price, quantity and order revenue related to one another?

Q.10 What KPIs best summarize product, customer, regional and channel performance?

Q.11 How could the same descriptive framework support inventory and merchandising decisions at real scale?

Q.12 How can this descriptive analysis lay the foundation for future predictive analytics such as demand forecasting?

Expected Outcome

The expected outcome is a structured analytical report in which every major visualization is supported by a clear description, observation and business insight, suitable for a BCA / Data Visualization academic submission.

How This Would Work at Real Scale

At real scale, a footwear retail analytics system could combine authorized point-of-sale and e-commerce data with inventory, customer-service and returns information. The descriptive framework used in this project could then be updated daily or in near real time.

Layer	Purpose	Tools
Inventory Management	Monitor model, size and material demand to reduce stock-outs and excess inventory. Warehouse-level availability can be compared with actual demand to prioritize replenishment.	Excel, Python
Product Planning	Identify models with consistently strong demand. New launches can be compared with previous models using units, revenue and average order value.	Python
Regional Planning	Compare validated country-level demand to decide where new distribution hubs or partnerships should be positioned.	Excel, Python
Marketing & Promotions	Evaluate campaigns by comparing units, revenue and basket size before, during and after promotional periods.	Python
Customer Experience	Combine payment preferences and order-value patterns to streamline checkout and reduce friction across channels.	Java, Python
Management Dashboards	Create dashboards showing KPIs such as revenue, units, AOV, size/material mix and regional performance so managers can monitor changes quickly.	Excel, Python

Business Insight

A real-time analytics system would allow the business to move from simple historical reporting toward continuous decision support, replenishing high-demand models faster and addressing regional or channel differences before they become larger operational problems.

Analytics Stack

This section documents the basic tools and technologies used to prepare, process and visualise the dataset for this project.

Tools & Technologies Used

Layer	Tool	Role in This Project
Data Collection / Staging	Microsoft Excel	Raw transaction records reviewed, cleaned and spot-checked (pivot tables, sorting, sanity checks) before analysis
Data Processing	Java	Basic data validation and preprocessing logic (parsing/formatting records, simple aggregation checks) prior to analysis
Analysis & Visualization	Python (Pandas, Matplotlib)	Core descriptive analytics — aggregation, KPI calculation, correlation analysis and chart generation
Development Environment	Jupyter Notebook	Interactive analysis and iterative chart building

KPI

- Primary preprocessing / validation language: Java
- Primary analysis & visualization language: Python (Pandas, Matplotlib)
- Primary staging / review tool: Microsoft Excel

Note on Stack Choice

Excel was used for lightweight initial data review since it is fast for eyeballing 100 records and building quick pivot summaries. Java handled basic preprocessing and validation logic, reinforcing data-structure fundamentals from coursework. Python (Pandas + Matplotlib) carried the core descriptive analytics workload — aggregation, correlation analysis and chart generation — since it is the standard toolchain for this kind of tabular analysis.

Business Insight

Documenting the analytics stack clarifies reproducibility: another analyst with Excel, Java and Python could reproduce the cleaning, validation and visualization steps used in this report using the same layered approach.

1. Top-Selling Shoe Models

Figure Description

The figure presents the ranking of US Polo shoe models by total gross revenue and by total pairs sold across the analysis period. The X-axis represents each model, while the Y-axis represents revenue or units. Comparing both views separates dollar-value leadership from physical-volume leadership.

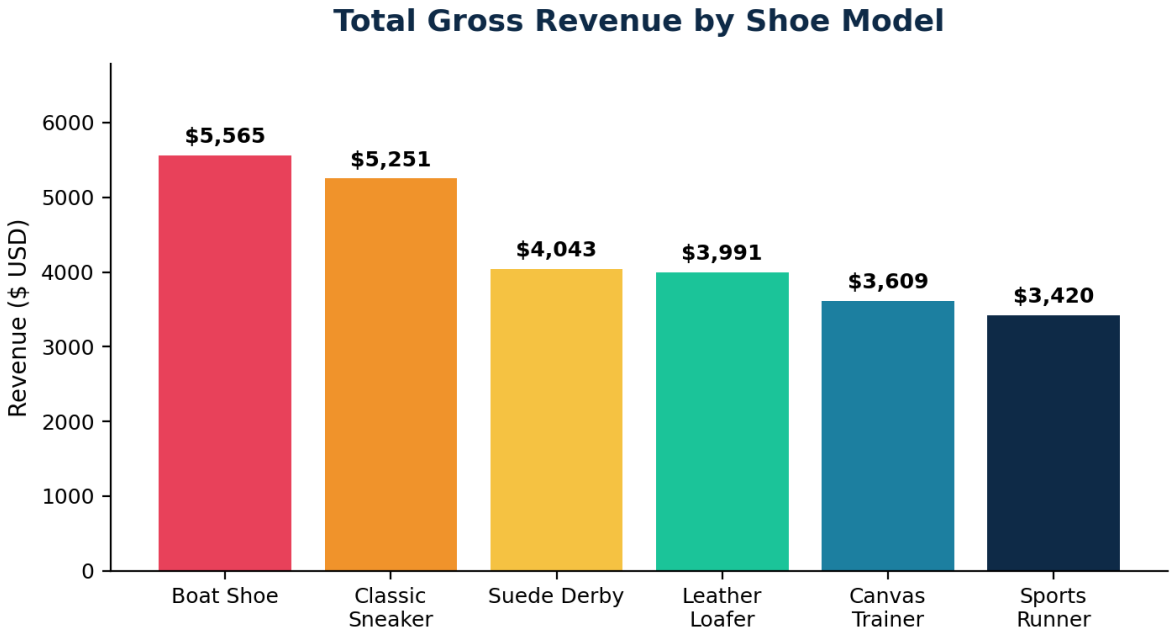


Figure 1a. Gross Revenue Contribution ($ USD) by Shoe Model.

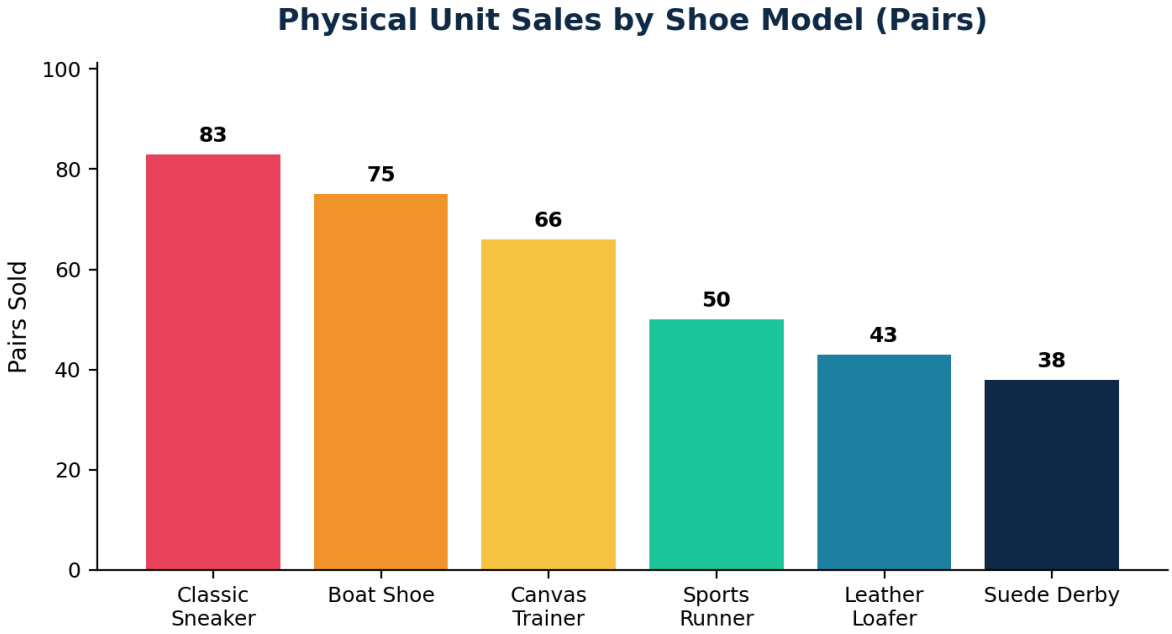


Figure 1b. Physical Unit Sales Volume by Shoe Model.

KPI

- Best-selling model by revenue: Boat Shoe — $5,564.98 (21.50% share)
- Best-selling model by volume: Classic Sneaker — 83 pairs
- Combined revenue of top 2 models: $10,815.54 (41.79% of total)
- Total simulated revenue: $25,879.10 across 355 pairs

Observation

Boat Shoe recorded the highest gross revenue at $5,564.98 across 21 orders, followed by Classic Sneaker at $5,250.56 across 24 orders. However, Classic Sneaker led in physical pair volume with 83 pairs sold — a divergence between dollar-value leadership and unit-volume leadership. The remaining models (Suede Derby, Leather Loafer, Canvas Trainer, Sports Runner) contributed comparatively smaller, though still meaningful, shares of total revenue.

Business Insight

High-performing models should receive priority in inventory replenishment and merchandising visibility. Classic Sneaker's high pair volume at a lower average order value suggests it is favoured for repeat, multi-pair purchasing and is a strong candidate for bundle or multi-buy promotions, while Boat Shoe's higher revenue concentration supports premium positioning.

Additional Interpretation

Model rankings alone do not explain why a product performs well. A stronger analysis would combine model demand with price, material, size tier and geographic market.

2. Monthly Sales Trend

Figure Description

The figure shows the monthly gross revenue trend from January to May 2025. The X-axis represents the sales month and the Y-axis represents total revenue for that month. The line shows how revenue activity changed across the recorded period.

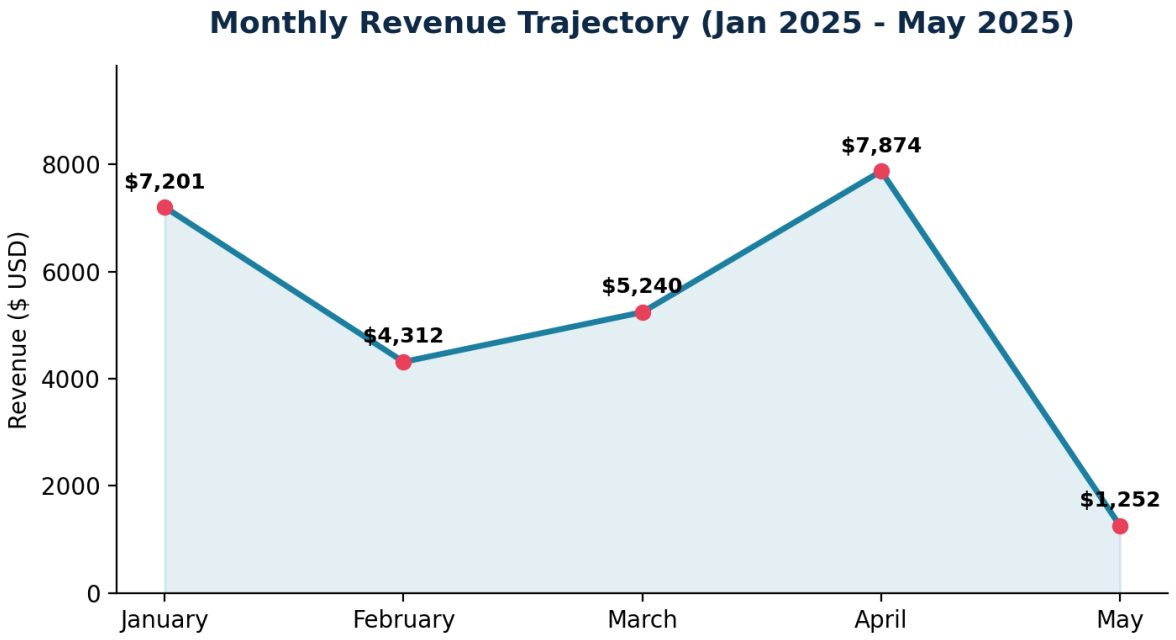


Figure 2. Monthly Sales Trend.

KPI

- Average monthly revenue: $5,175.82
- Highest-revenue month: April 2025 — $7,874.27 (30.43% share)
- Lowest-revenue month: May 2025 (partial) — $1,251.98

Observation

Revenue peaked in April 2025 at $7,874.27 across 24 orders and 99 pairs. January was also strong at $7,200.77, while February dipped to $4,312.18 before recovering in March. May's low value reflects the dataset's collection boundary ending on 1 May 2025 — only 5 orders were logged that month, so the apparent decline is a structural artifact rather than a genuine demand contraction.

Business Insight

Identifying high-revenue months can help distributors plan inventory positioning and time promotional campaigns. April's peak suggests a seasonal or promotional driver worth investigating with additional data, while the May figure should be excluded from any month-over-month demand comparison.

Additional Interpretation

The five-month window demonstrates the analytical method but is too short to establish genuine seasonal patterns; a full 12-month dataset would be needed to confirm seasonality.

3. Average Order Value & Distribution

Figure Description

The figure presents the number of orders falling into each $150 USD revenue bracket. The X-axis represents the order-value bracket and the Y-axis represents the number of orders. This distribution shows whether sales are dominated by single-pair shoppers or bulk buyers.

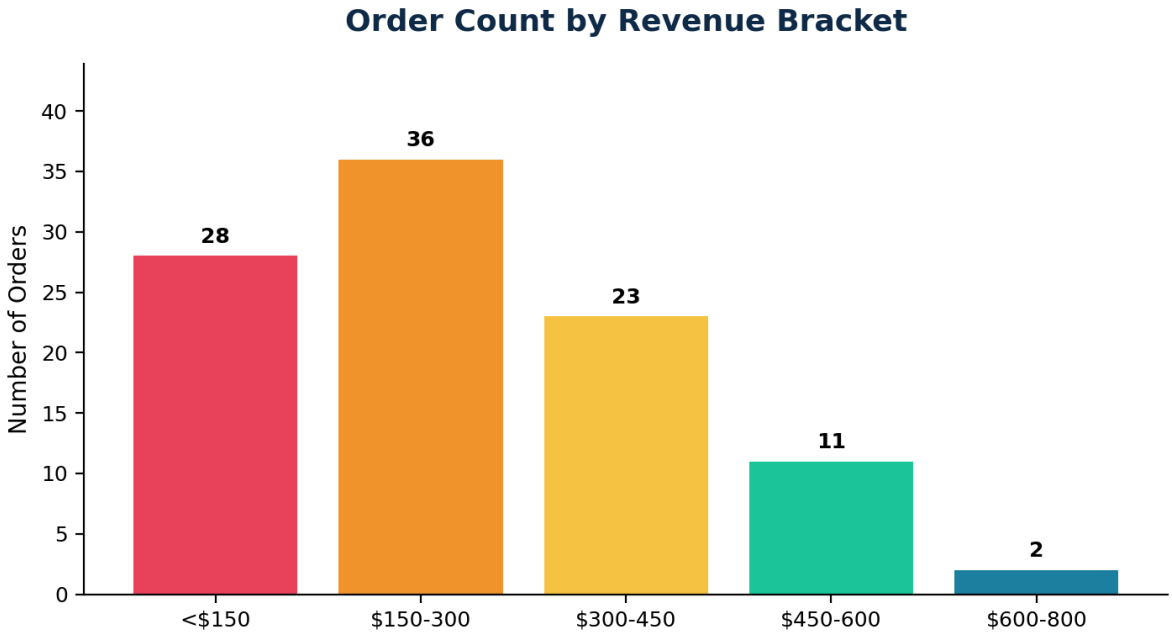


Figure 3. Average Order Value & Distribution.

KPI

- Overall Average Order Value (AOV): $258.79
- Median order value: $219.42
- Order value range: $44.29 – $734.64
- Average basket size: 3.55 pairs per order

Observation

Most orders (64%) fall under $300, indicating that single-pair and small-boutique purchases dominate transaction volume. A smaller number of high-value orders in the $450–800 range pull the mean ($258.79) above the median ($219.42), producing a right-leaning distribution typical of a retail dataset with occasional bulk purchases.

Business Insight

Checkout flows should support both ends of the distribution: streamlined single-pair purchase paths for the majority of customers, and dedicated bulk-order features — tiered discounts, invoicing, reorder scheduling — to convert the smaller but higher-value bracket of bulk buyers.

Additional Interpretation

The gap between mean and median should be monitored over a larger sample; a few large orders can meaningfully shift the average even when they are not representative of typical customer behaviour.

4. Size Tier Allocation

Figure Description

The figure presents total pairs sold by size tier. The X-axis represents the three size tiers and the Y-axis represents pairs sold. Size analysis is important for footwear because assortment availability by size directly affects whether a customer can complete a purchase.

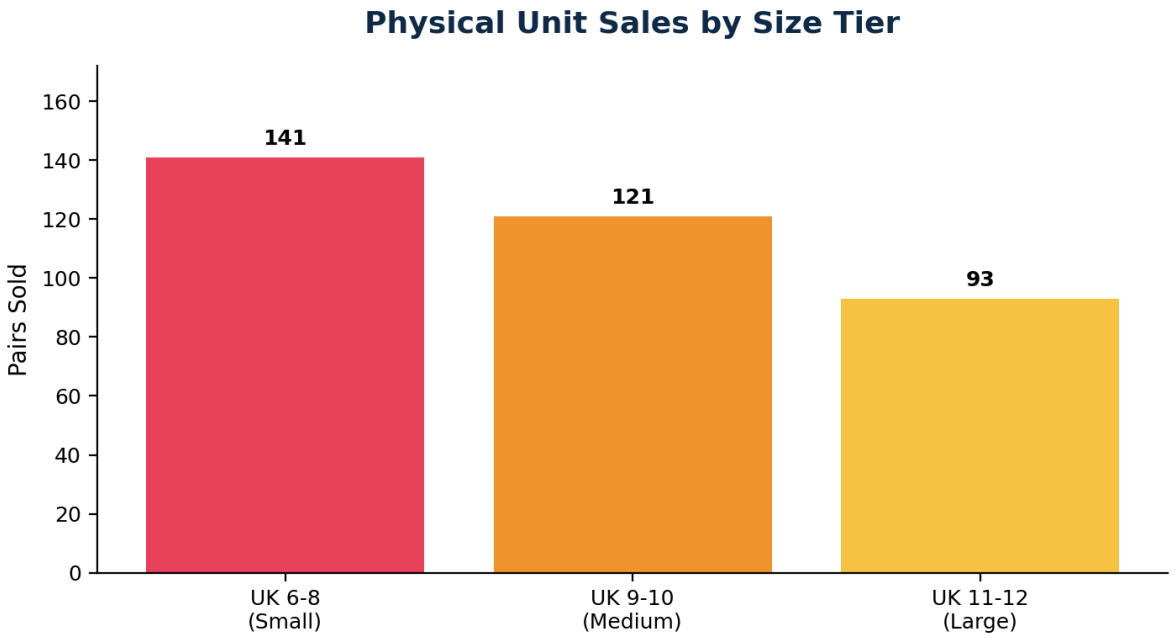


Figure 4. Size Tier Allocation.

KPI

- Most-demanded tier: UK 6-8 (Small) — 141 pairs (39.72% unit share)
- Highest average order value: UK 11-12 (Large) — $299.62
- Tiers analysed: 3

Observation

UK 6-8 (Small) is the clear market favourite by unit volume, followed closely by UK 9-10 (Medium). UK 11-12 (Large) sold fewer pairs overall but carried the highest average order value, suggesting large-size buyers purchase in higher-value baskets even though they are less frequent.

Business Insight

Production and stocking should be weighted roughly in proportion to observed unit share (Small 39.72%, Medium 34.08%, Large 26.20%), while the higher average order value of the Large tier makes it a good candidate for premium or multi-pair promotions.

Additional Interpretation

Size demand should be measured over a longer period and adjusted for stock availability before being used for procurement planning.

5. Material Type Share

Figure Description

The figure compares total gross revenue across the five recorded upper-material types. The X-axis represents material type and the Y-axis represents total revenue. The chart shows which materials contribute the largest share of merchandising revenue.

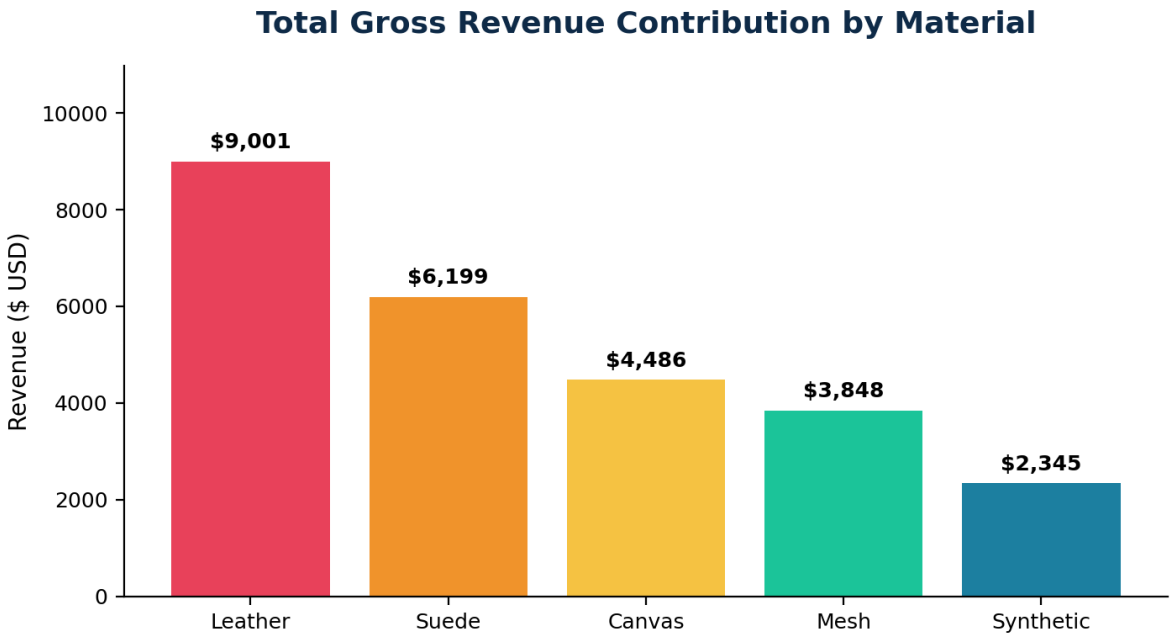


Figure 5. Material Type Share.

KPI

- Highest-revenue material: Leather — $9,000.51 (34.78% share)
- Highest average order value: Suede — $364.66
- Lowest-revenue material: Synthetic — $2,345.01 (9.06% share)

Observation

Leather generated the highest total revenue across 27 orders and 111 pairs, while Suede — despite fewer orders — produced the highest average revenue per order ($364.66), indicating that premium-material buyers disproportionately drive higher-value baskets. Synthetic recorded both the fewest orders and the lowest revenue share.

Business Insight

Leather and Suede should be featured prominently in merchandising and homepage placement to capture premium order values, while Synthetic and Mesh may benefit from bundling or price-positioning review.

Additional Interpretation

Material popularity is a descriptive measure and should be combined with profitability and stock cost before final merchandising decisions.

6. Regional Demand Distribution

Figure Description

The figure compares total revenue across the six countries represented in the dataset. The X-axis represents country and the Y-axis represents revenue. The chart is intended to show geographic concentration of demand.

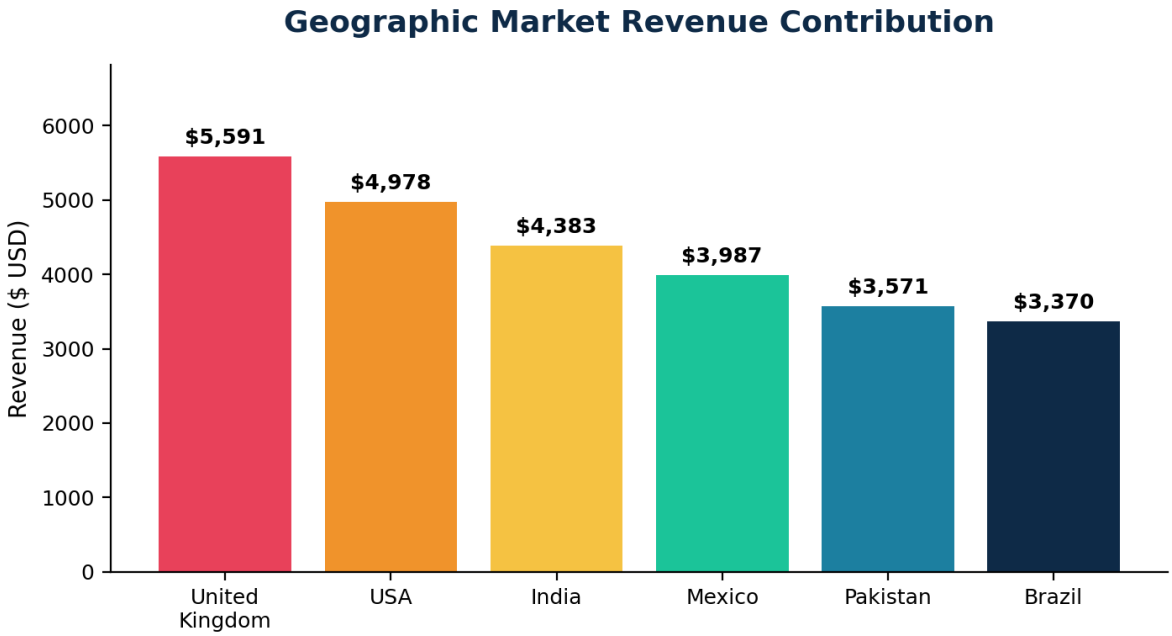


Figure 6. Regional Demand Distribution.

KPI

- Highest-revenue market: United Kingdom — $5,590.76 (21.60% share)
- Second-highest market: USA — $4,977.55 (19.23% share)
- Markets represented: 6

Observation

United Kingdom and USA together account for $10,568.31 (40.83%) of global revenue. India, Mexico, Pakistan and Brazil each contribute more modest, fairly even shares (13–17%), indicating a geographically diversified — though UK/USA-weighted — customer base.

Business Insight

Logistics and inventory allocation should prioritise the United Kingdom and USA while treating India, Mexico, Pakistan and Brazil as promising markets for targeted regional promotions or localized financing programs.

Additional Interpretation

These figures represent this academic dataset only and should not be interpreted as official regional sales for the brand.

7. Revenue Concentration

Figure Description

The table below summarizes revenue concentration across the two leading shoe models compared with the remaining four models, illustrating how much of total revenue is concentrated in the top-performing tier.

Tier	Models Included	Pairs Sold	Revenue ($)	Share (%)
Household Tier	Boat Shoe, Classic Sneaker	158	$10,815.54	41.79%
Specialty & Premium Tier	Suede Derby, Leather Loafer, Canvas Trainer, Sports Runner	197	$15,063.56	58.21%
Total	All 6 Models	355	$25,879.10	100.00%

KPI

- Top-2-model share of orders: 45.00%
- Top-2-model share of pair volume: 44.51%
- Top-2-model share of revenue: 41.79%

Observation

Boat Shoe and Classic Sneaker together account for 45.00% of orders, 44.51% of pair volume, and 41.79% of revenue, forming a clear “household” tier. The remaining four models make up a broader specialty tier that together generates a majority of revenue (58.21%) but from a wider, less-concentrated base.

Business Insight

The household tier should be treated as the primary, highest-frequency revenue anchor and protected against stock-outs, while the specialty tier provides style variety and should be positioned to capture premium and repeat-buyer demand across mature markets.

8. Customer Payment Preferences

Figure Description

The figure presents the number of orders by checkout payment method. The X-axis represents the payment channel and the Y-axis represents the number of orders. This comparison shows how customers prefer to settle transactions.

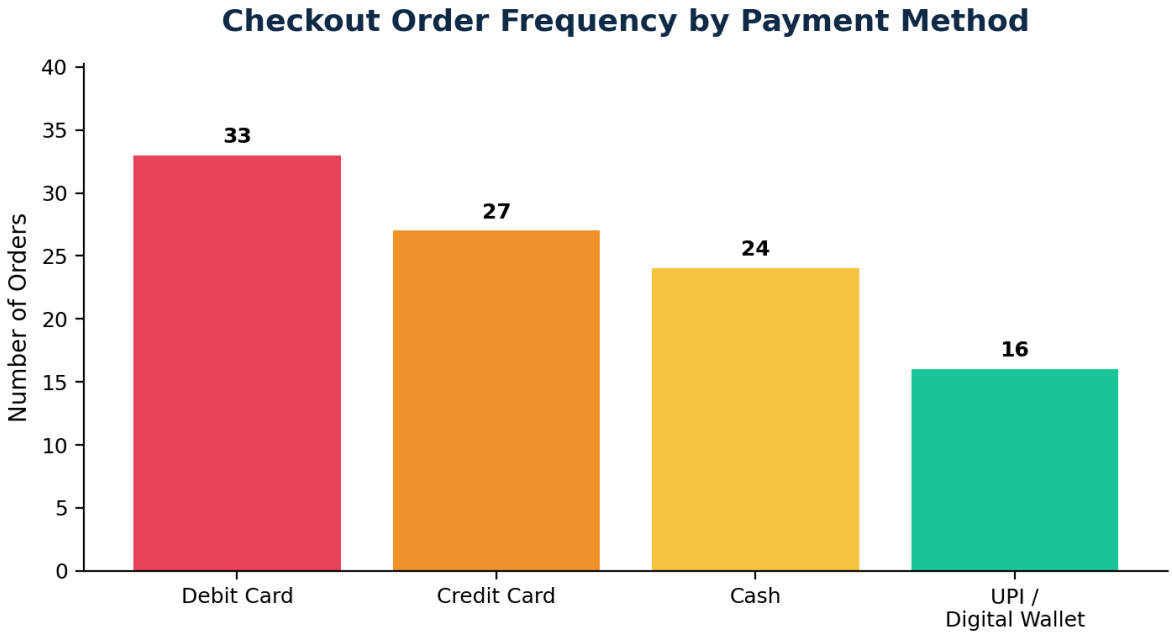


Figure 8. Customer Payment Preferences.

KPI

- Leading payment method: Debit Card — 33 orders (33.00% share)
- Combined card-rail share (Debit + Credit): 60.00% of orders
- Digital wallet (UPI) share: 16.00% of orders

Observation

Debit Card is the primary checkout method (33 orders, $8,251.46 revenue), followed by Credit Card (27 orders, $7,682.18). Combined, traditional card rails account for 60% of orders and 61.57% of revenue. Cash remains meaningful at 24 orders, while UPI/digital wallet — the newest channel — already captures 16% of orders.

Business Insight

Checkout should continue to support diverse payment rails, with particular attention to expanding digital-wallet and Buy-Now-Pay-Later options, which could reduce friction for smaller or newer retail accounts without displacing the dominant card-based channels.

9. Price & Basket Size Relationships

Figure Description

The table below summarizes the statistical relationships between unit price, basket quantity and total order revenue, clarifying which factor most strongly drives order-level revenue.

Variable Pair	Correlation (r)	Strength	Interpretation
Quantity vs. Revenue	+0.86	Strong Positive	Basket pair count is the primary revenue driver
Unit Price vs. Revenue	+0.58	Moderate Positive	Higher prices raise order value, but less than quantity
Unit Price vs. Quantity	+0.14	Weak / Neutral	Higher prices do not meaningfully shrink basket size

KPI

- Quantity–Revenue correlation: r = +0.86 (strong)
- Price–Revenue correlation: r = +0.58 (moderate)
- Price range observed: $36.83 – $122.44 per pair

Observation

Basket quantity correlates far more strongly with order revenue (r = +0.86) than unit price does (r = +0.58). Price and quantity themselves are only weakly related (r = +0.14), meaning higher-priced pairs are not being bought in meaningfully smaller baskets.

Business Insight

Because basket size is the dominant lever on order revenue, promotions and merchandising that encourage multi-pair purchases (bundles, tiered discounts, free-shipping thresholds) are likely to have a larger effect on revenue than price adjustments alone.

Additional Interpretation

Correlation reflects association within this dataset, not a controlled causal experiment; other unobserved factors could influence both price and basket size.

Summary of Key Metrics

The following KPI table consolidates the main results of the dataset into a single management-style view.

Objective	Key Metric	Result
Product performance	Top model by revenue	Boat Shoe
Product performance	Top model revenue	$5,564.98
Product performance	Top model by volume	Classic Sneaker (83 pairs)
Monthly trend	Peak revenue month	April 2025 ($7,874.27)
Order value	Average Order Value (AOV)	$258.79
Order value	Median order value	$219.42
Size tier	Most-demanded tier	UK 6-8 (Small)
Material	Top material by revenue	Leather ($9,000.51)
Regional	Top market	United Kingdom ($5,590.76)
Payment	Top payment method	Debit Card (33 orders)
Concentration	Top-2-model revenue share	41.79%
Revenue	Total gross revenue	$25,879.10
Volume	Total pairs sold	355
Correlation	Quantity–Revenue r	+0.86

KPI Interpretation

The KPI summary shows that product demand, order value, size, material, geography, payment preference and revenue concentration can all be measured directly from transaction-level data. In a real retail environment, these measures could be placed on a dashboard and monitored regularly to identify changes in demand and support faster decisions.

Important Academic Limitation

Every value in this KPI table is calculated from the academic project dataset (100 orders, 355 pairs, January–May 2025). It must not be presented as official U.S. Polo Assn. / USPA Global Licensing sales or customer statistics.

Next Steps: Descriptive → Predictive Analytics

The descriptive analysis performed in this project provides a foundation for future predictive analytics. Descriptive analytics answers questions such as which model sold the most and which market generated more revenue. Predictive analytics can then use historical patterns to estimate what may happen next.

Possible Predictive Applications

- Forecast future pair demand by model, size and country using historical sales patterns.
- Predict which incoming orders are likely to exceed a high-value threshold.
- Estimate customer lifetime spend for repeat retail accounts.
- Flag unusual demand spikes that could indicate stock-out risk.
- Recommend models or materials based on previous purchases and market.
- Estimate expected revenue under different promotional or bundling scenarios.

Requirement for Real Implementation

Predictive models require larger, validated datasets, appropriate features, model evaluation and responsible handling of customer information. The current 100-order dataset is suitable for demonstrating descriptive analysis but is not sufficient to train a reliable production forecasting model.

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Dataset Citation

US Polo Shoes Sales Transaction Dataset (100 orders, 355 pairs, Jan–May 2025): academic project dataset recording Order ID, Customer Name, Country, Shoe Model, Size Tier, Material, Quantity, Price, Sale Date, Payment Method, and computed Revenue.

Source-use Note

Official brand sources are used only for general brand/product context. All transaction-level sales, order-value, regional, payment and KPI results in this report are generated from the academic project dataset.

Appendix

Appendix A — Dataset Schema

Field	Type	Sample Value	Description
Order_ID	Integer	1 … 100	Unique transaction primary key
Country	Categorical	USA	Delivery destination (6 markets)
Shoe_Model	Categorical	Boat Shoe	Footwear model (6 models)
Size_Tier	Categorical	UK 9-10 (Medium)	Size tier
Material	Categorical	Leather	Upper material (5 types)
Quantity	Integer	3	Pairs purchased (1–6)
Price	Float	72.90	Unit price (USD)
Payment_Method	Categorical	Debit Card	Settlement channel
Revenue	Float	258.79	Quantity × Price

Appendix B — Python / Pandas Preparation

```
df['Revenue'] = df['Quantity'] * df['Price']
```

```
aov = df['Revenue'].sum() / df['Order_ID'].nunique()
```

Appendix C — Academic Limitation

The dataset represents 100 academic project orders and cannot represent actual U.S. Polo Assn. sales, customers, revenue or official market performance. This appendix exists to make the analysis reproducible.